FIFA PLAYER MARKET VALUE PREDICTION

Machine Learning Project Report

Dataset: FIFA Player Statistics (5,682 Players, 41 Features)

Target Variable: Player Market Value (USD)

Models Used: Linear Regression · Random Forest · Tuned Random Forest

Best Model R^2 Score: 0.9106 (Tuned Random Forest)

Key Technique: PCA, GridSearchCV Hyperparameter Tuning

Report Sections: 12 Major Sections | 30+ Pages

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1. Problem Statement

The football industry has evolved into a multi-billion-dollar global business where accurate player valuation is critical for clubs, agents, scouts, and investors. Determining a footballer's fair market value, however, remains a complex challenge influenced by a vast array of physical, technical, and contextual attributes.

Traditional approaches to player valuation rely heavily on subjective expert opinion, recent match performance, and agent negotiations. These methods are inherently inconsistent, prone to bias, and lack the scalability required in modern football analytics. There is a growing need for a data-driven, transparent, and reproducible framework that can estimate player market value with high accuracy.

This project addresses that gap by constructing a machine learning pipeline using the FIFA player statistics dataset. The primary objective is to train regression models that can accurately predict a player's market value (in USD) from their in-game attributes and demographic information. The study further investigates the effect of dimensionality reduction via PCA and the impact of hyperparameter optimisation on predictive accuracy.

1.1 Research Questions

- Which player attributes are the strongest predictors of market value?
- Can machine learning models outperform simple linear baselines for this regression task?
- Does PCA-based dimensionality reduction improve or degrade model performance?
- What is the effect of hyperparameter tuning on Random Forest regression accuracy?
- Are there non-linear relationships between physical/skill attributes and market value?

1.2 Significance of the Study

Accurate market valuation has direct implications for transfer fee negotiations, contract renewals, squad planning, and fantasy sports platforms. A reliable predictive model enables smaller clubs to identify undervalued talent ('moneyball' strategy), assists agents in benchmarking fair compensation, and provides transparency to fans and analysts.

> **INSIGHT:** The highest valued player in the dataset is Kylian Mbappé at $153,500,000 USD, while the minimum value recorded is $400 USD — a range spanning over 8 orders of magnitude, underlining the complexity of this prediction task.

2. Business Objectives

The project is framed around concrete, measurable business objectives aligned with the needs of football clubs, sports analytics firms, and transfer market platforms. These objectives guided all design decisions throughout the machine learning pipeline.

2.1 Primary Objectives

ID	Objective	Success Criterion
BO-1	Predict Market Value	Build a regression model that predicts FIFA player market value with $R^2 \geq 0.85$ on held-out test data.
BO-2	Feature Identification	Identify the top 10 player attributes that most significantly drive market value.
BO-3	Model Comparison	Compare Linear Regression vs. Random Forest to understand the trade-off between interpretability and accuracy.
BO-4	Dimensionality Study	Evaluate whether PCA-based feature compression preserves predictive performance.
BO-5	Hyperparameter Tuning	Optimise the best-performing model via GridSearchCV to achieve maximum generalisation.

2.2 Secondary Objectives

- Develop a reproducible data preprocessing pipeline applicable to similar sports datasets.
- Provide actionable insights on player development and recruitment strategies for mid-tier clubs.
- Quantify the relationship between age and market value to assist in youth development investment decisions.
- Demonstrate the practical utility of PCA for high-dimensional tabular sports data.
- Establish a baseline framework that can be extended to other sports or seasons.

2.3 Stakeholder Mapping

Stakeholder	Context	Use of Model
Football Clubs	Transfer targets, squad budgeting	Predict value of transfer targets
Sports Agents	Contract negotiations	Benchmark fair market value for clients
Fantasy Platforms	Player pricing algorithms	Automate value assignment at scale
Sports Analysts	Academic & media research	Quantify attribute-value relationships
Scouts & Recruiters	Youth talent identification	Identify undervalued emerging players

3. Dataset Overview

3.1 Dataset Description

The dataset used in this project is derived from FIFA player statistics and contains comprehensive attribute profiles for professional football players. The data captures in-game ratings that reflect each player's real-world abilities as modelled by the FIFA video game franchise, supplemented with demographic and club information.

Attribute	Value
Source	FIFA Player Statistics Database
File Format	CSV (Comma-Separated Values), Latin-1 Encoding
Total Records	5,682 Players
Total Features	41 Columns (including target variable)
Target Variable	value (Player Market Value in USD)
Value Range	$400 to $153,500,000 USD
Mean Market Value	$2,271,115 USD
Median Market Value	$77,500 USD
Age Range	17 to 41 years
Number of Countries	131+ nationalities
Number of Clubs	500+ professional clubs
Missing Data	Only 'marking' column (100% null)

3.2 Feature Categories

The 41 features can be grouped into five logical categories, each capturing a distinct dimension of player ability or identity:

Category	# Features	Features Included
Demographic	4	player, country, age, height, weight, club
Technical Skills	15	ball_control, dribbling, crossing, short_pass, long_pass, curve, fk_acc, penalties, volleys, finishing, long_shots, shot_power, att_position, vision, composure
Defensive Skills	5	marking (all null), slide_tackle, stand_tackle, aggression, interceptions
Physical Attributes	7	acceleration, stamina, strength, balance, sprint_speed, agility, jumping
Heading & Set Piece	2	heading, reactions
Goalkeeper Stats	5	gk_positioning, gk_diving, gk_handling, gk_kicking, gk_reflexes
Target Variable	1	value (market value in USD)

3.3 Target Variable: Market Value

The target variable 'value' represents the player's estimated market value in USD, stored as a formatted string (e.g., '$1.400.000'). This required custom parsing during preprocessing. The distribution of raw market values is highly right-skewed, which is typical of salary and financial datasets. The top 5% of players account for a disproportionate share of total market value, reflecting the 'superstar economy' in football.

Percentile Range	Value (USD)
Bottom 25% (Q1)	≤ $35,000
25th – 50th Percentile	$35,000 – $77,500
50th – 75th Percentile	$77,500 – $1,600,000
Top 25% (Q4)	> $1,600,000
Maximum (Kylian Mbappé)	$153,500,000
Standard Deviation	$7,616,233

> **PREPROCESSING NOTE:** Due to the extreme right skew of market values, a log1p transformation (log(1 + x)) was applied during preprocessing. This transformation compressed the scale, improved model convergence, and reduced the influence of extreme outliers such as Mbappé ($153.5M) on regression coefficients.

4. Data Quality Assessment

A thorough data quality assessment was performed before any modelling. This phase ensures that the data is clean, consistent, and ready for machine learning. The assessment covered missing values, duplicates, encoding issues, outliers, and feature relevance.

4.1 Missing Value Analysis

The dataset was inspected for null values across all 41 columns. The findings revealed a single column with complete data loss and the rest with no missing values at all:

Column	Missing Count	% Missing	Action Taken
marking	5,682 / 5,682	100.0%	Entirely null — dropped from modelling (retained structurally but excluded)
All other 39 features	0 / 5,682	0.0%	Complete data — no imputation required

The 'marking' column, which represents a player's defensive marking ability, was entirely null across all 5,682 records. This is likely due to the nature of the dataset — the FIFA game may not have assigned 'marking' ratings for all player types, or this attribute was deprecated in the version of the data collected. Since the median of an all-null column remains NaN even after median imputation, this column was explicitly removed prior to PCA and Linear Regression using index-based deletion.

4.2 Duplicate Records

Duplicate detection was performed using pandas' .duplicated() function. A small number of duplicate entries were identified and removed. The presence of duplicates may be attributed to data aggregation from multiple FIFA editions or regions.

- Duplicates detected: 3 rows
- Action: Removed using df.drop_duplicates(inplace=True)
- Final dataset size after deduplication: 5,679 rows

4.3 Data Type Inconsistencies

Several columns required type conversion before analysis:

Column	Original Type	Final Type	Transformation
value	object (string)	float64	Regex parsing to strip '$', '.', and spaces, then cast to float
country	object (string)	int64 (encoded)	Label Encoding applied during preprocessing
club	object (string)	int64 (encoded)	Label Encoding applied during preprocessing
All skill ratings	int64	float64	Promoted during StandardScaler transformation

4.4 Outlier Detection

Outlier detection was performed on the log-transformed 'value' column using the Interquartile Range (IQR) method. This is the standard approach for continuous regression targets:

- Q1 (25th percentile): computed from log-transformed values
- Q3 (75th percentile): computed from log-transformed values
- IQR = Q3 – Q1
- Lower bound = Q1 – 1.5 × IQR | Upper bound = Q3 + 1.5 × IQR
- Rows outside these bounds were removed to eliminate extreme value distortions

This step removes statistically anomalous records that could disproportionately influence model coefficients, particularly for linear regression. The retained dataset remains representative of the majority of professional players while reducing noise.

4.5 Encoding Consistency

The categorical variables 'country' and 'club' were encoded using scikit-learn's LabelEncoder. This assigns a unique integer to each category, making the data compatible with numerical ML algorithms. Key observations:

- 131+ unique nationalities encoded to integer values 0–130+
- 500+ unique clubs encoded to integer values 0–499+

> **LIMITATION:** LabelEncoder introduces an implicit ordinal relationship between categories (e.g., club 0 < club 1). For production systems, One-Hot Encoding or Target Encoding would be more appropriate. However, for tree-based models like Random Forest, LabelEncoder is generally acceptable since splits can correct for false ordinality.

5. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was conducted to understand the structure, distribution, and relationships within the dataset before model building. EDA informed key preprocessing decisions and provided domain-relevant insights about FIFA player characteristics.

5.1 Univariate Analysis

5.1.1 Market Value Distribution

The raw market value distribution is heavily right-skewed, with the vast majority of players valued below $5 million and a long tail extending toward elite players valued above $100 million. Key statistics are summarised below:

Statistic	Value
Minimum Value	$400
Maximum Value	$153,500,000 (Kylian Mbappé)
Mean Value	$2,271,115
Median Value	$77,500
Standard Deviation	$7,616,233
Distribution Shape	Highly right-skewed (positive skew)
Log-transformed Distribution	Approximately normal after log1p

After applying the log1p transformation, the distribution becomes approximately bell-shaped, enabling linear models to perform effectively. The log-transformed values range from approximately 6.0 to 18.85.

5.1.2 Age Distribution

The age of players in the dataset ranges from 17 to 41 years, with the distribution peaking in the 22–26 age bracket — corresponding to the prime competitive age for most outfield footballers. The distribution is slightly right-skewed due to a smaller pool of players aged 35 and above who remain active at professional level.

Age Group	Count	% of Dataset
17–20 (Youth)	590 players	10.4%
21–25 (Rising)	1,996 players	35.1%
26–30 (Prime)	1,754 players	30.9%
31–35 (Experienced)	1,136 players	20.0%
36+ (Veteran)	206 players	3.6%

5.1.3 Physical Attributes

Player heights range from 156 cm to 204 cm with a mean of 181.7 cm, consistent with the physical profile of professional footballers. Weight averages 75.3 kg with a standard deviation of 7.0 kg. These attributes show moderate correlation with certain positional skills such as heading and strength.

Feature	Min	Max	Mean	Std Dev
Height (cm)	156	204	181.7	6.8
Weight (kg)	54	102	75.3	7.0
Age (years)	17	41	26.3	4.7

5.2 Bivariate Analysis

5.2.1 Correlation with Market Value

A correlation heatmap (seaborn heatmap with coolwarm palette) was generated to visualise the pairwise correlation between all numeric features and the target variable. Key findings:

Feature	Correlation with Value	Interpretation
reactions	+0.76	Highest correlation — reflects overall player quality perception
ball_control	+0.68	Strong predictor of technical proficiency
composure	+0.64	Mental attributes strongly valued in top-tier players
vision	+0.62	Playmaking ability is rewarded in the transfer market
short_pass	+0.61	Possession-based football demands precise passing
dribbling	+0.59	Closely associated with attacking roles and high value
age	+0.41	Moderate positive — prime-age players attract higher fees
gk_diving	-0.38	Goalkeepers have very different value profiles
strength	+0.22	Physical power has modest correlation with value
jumping	+0.18	Weaker predictor — specific to aerial specialists

5.2.2 Reactions vs. Market Value

A scatter plot of 'reactions' versus log-transformed 'value' reveals a strong positive linear trend, confirming reactions as the most powerful individual predictor. Players with reactions scores above 75 tend to cluster at higher value brackets, while those below 60 are predominantly in the lower quartile of market value. The feature importance analysis from Random Forest later confirms reactions as the dominant predictor (59.05%).

5.2.3 Ball Control vs. Market Value

Ball control shows a similar but noisier positive relationship with value. The scatter exhibits more variance at intermediate skill levels (50–70), suggesting that ball control alone is insufficient — it must be combined with other attributes to command a premium value. This validates the multi-feature approach used in the machine learning models.

5.3 Multivariate Analysis

Beyond pairwise correlations, several multivariate patterns were observed:

- Goalkeeper attributes (gk_diving, gk_reflexes, gk_handling) form a tight cluster of inter-correlated features, distinct from outfield player attributes.
- Technical skill attributes (ball_control, dribbling, short_pass, vision, composure) are highly inter-correlated ($r > 0.6$), suggesting multicollinearity that PCA can address.
- Physical attributes (strength, jumping, heading) correlate positively with each other but weakly with market value, implying physical power is less commercially valued than technical skill.
- Players aged 23–29 in the prime bracket command the highest average values across all skill levels, confirming age as a moderating variable.

5.4 Geographic and Club Distribution

Country	Players	% of Dataset	Note
England	516	9.1%	Largest representation — Premier League depth
Germany	390	6.9%	Strong Bundesliga presence
Spain	356	6.3%	La Liga and lower-tier clubs
France	309	5.4%	Ligue 1 and feeder leagues
Argentina	302	5.3%	South American talent pipeline
Brazil	223	3.9%	Globally exported talent
Italy	186	3.3%	Serie A squads
Netherlands	146	2.6%	Ajax academy pipeline

6. Feature Engineering & Preprocessing

A structured preprocessing pipeline was designed and implemented to transform raw data into a format suitable for machine learning. Each step is documented with its rationale, implementation, and impact on the dataset.

6.1 Preprocessing Pipeline Overview

Step	Operation	Details
Step 1	Data Loading	pd.read_csv with latin1 encoding to handle special characters in player names
Step 2	Missing Value Imputation	Median imputation for numerical columns; mode imputation for categorical (club, country)
Step 3	Duplicate Removal	df.drop_duplicates() — removed 3 duplicate rows
Step 4	Feature Dropping	Removed 'player' column (identifier, not a predictive feature)
Step 5	Target Parsing	Regex-cleaned value string → float conversion
Step 6	Log Transformation	np.log1p(value) to normalise skewed target
Step 7	Outlier Removal	IQR-based filtering on log-transformed target
Step 8	Categorical Encoding	LabelEncoder for 'country' and 'club'
Step 9	Train-Test Split	80/20 split, random_state=42 for reproducibility
Step 10	Feature Scaling	StandardScaler — fit on train, transform both train and test
Step 11	Null Column Removal	'marking' column excluded for LR and PCA pipelines

6.2 Target Variable Transformation

The 'value' column required the most complex transformation. The raw string format used European number notation (periods as thousand separators) and required careful regex handling:

- Raw format example: '$1.400.000' representing $1,400,000 USD
- Regex applied: replace('[$,.]', '', regex=True) to strip currency symbols and separators
- Cast to float and then log1p applied: value = np.log1p(numeric_value)

The log1p transformation has several advantages: it handles zero values gracefully, compresses the large dynamic range of values, makes the distribution more Gaussian (favouring linear models), and reduces the influence of extreme outliers on loss functions.

6.3 Feature Scaling

StandardScaler was applied to all numerical features before Linear Regression and PCA. This centres each feature to zero mean and unit variance, ensuring that features with large absolute ranges (e.g., weight in kg) do not dominate features with smaller ranges (e.g., agility scores 0–100).

> **DATA LEAKAGE PREVENTION:** StandardScaler was fitted exclusively on X_train and then used to transform X_test. Fitting the scaler on the full dataset (including test) would constitute data leakage — a critical methodological error. This pipeline correctly avoids it.

6.4 Train-Test Split Strategy

Split	Rows	Proportion	Purpose
Training Set	4,543 rows	80%	Used for model fitting and GridSearchCV
Test Set	1,136 rows	20%	Held out — used only for final evaluation
Random State	42	—	Ensures reproducibility across all experiments
Stratification	None	—	Random split; regression target does not require stratification

6.5 Feature Selection Rationale

The 'player' column was dropped as it is a unique identifier with no predictive signal. The 'marking' column was excluded only from the Linear Regression and PCA pipelines because it remained fully null after imputation (the median of an all-null column is NaN). Random Forest, however, was able to handle this during training via the raw (unscaled) feature matrix without the marking column being explicitly removed, as the tree algorithm handles NaN through surrogate splits in some implementations.

7. Dimensionality Reduction (PCA)

Principal Component Analysis (PCA) is an unsupervised linear transformation technique that projects high-dimensional data onto a lower-dimensional subspace while maximising the retained variance. PCA was applied in this project to investigate whether a compressed feature representation could match the performance of the full feature set.

7.1 Theoretical Background

PCA works by computing the eigenvectors (principal components) of the feature covariance matrix. Each principal component is an orthogonal linear combination of the original features, ranked by the amount of variance it captures. The first principal component captures the most variance, with each subsequent component capturing progressively less.

Key advantages of PCA in this context:

- Addresses multicollinearity among technical skill attributes (e.g., dribbling and ball_control are highly correlated)
- Reduces dimensionality from 38 features to potentially fewer components without significant information loss
- Can improve the generalisation of linear models by removing redundant feature dimensions
- Enables visualisation of high-dimensional data in 2D or 3D space

7.2 PCA Implementation

PCA was applied using scikit-learn's PCA() with all components initially retained to generate the full variance plot. The 'marking' column (all null) was removed before applying PCA, reducing the input dimensionality from 39 to 38 features.

PCA Parameter	Value / Description
Input Features	38 (after excluding 'marking' column)
Standardisation	StandardScaler applied before PCA (mandatory for PCA)
PCA Method	Full SVD decomposition (sklearn default)
Components Computed	38 (all components)
Components for 90% Variance	10 principal components
Components for 95% Variance	17 principal components
Components for 99% Variance	~28 principal components
Variance in PC1	Largest single component — captures dominant variance axis

7.3 Explained Variance Analysis

The cumulative explained variance plot (generated via matplotlib) reveals that:

- 10 principal components are sufficient to capture 90% of the total data variance, representing a 74% reduction in dimensionality (38 → 10 features).
- 17 principal components retain 95% of variance — a 55% reduction.
- The first few components capture dominant variance structures, primarily related to the cluster of inter-correlated technical skill attributes.

This rapid accumulation of variance in the early components reflects the high degree of multicollinearity in the original feature set. Football skill attributes (dribbling, ball_control, passing, vision) tend to move together — players who excel in one technical area typically excel across multiple areas — creating correlated dimensions that PCA collapses.

7.4 Impact of PCA on Model Performance

Model Configuration	MAE	MSE	R^2 Score
Linear Regression (Full Features)	0.9269	1.3321	0.7516
Linear Regression with PCA	0.9269	1.3321	0.7516
Difference	0.0000	0.0000	0.0000

Remarkably, Linear Regression with PCA achieves identical performance to the full-feature Linear Regression. This is mathematically expected: when PCA is applied to all components before linear regression, it is equivalent to a rotation of the feature space — the predictive information content is fully preserved. The result would only differ if a reduced number of components (< 38) were used, in which case some variance would be discarded, potentially improving generalisation or reducing overfitting.

> **KEY FINDING:** The equivalence of LR and LR+PCA (all components) is a mathematical identity. In practice, PCA is most useful when applied with a reduced number of components (e.g., retaining only the top 10–17 components for 90–95% variance). Future work should explore this truncated PCA regime and its effect on test set generalisation.

8. Model Development

Two regression models were developed and evaluated: Linear Regression (as a baseline) and Random Forest Regressor (as the primary model). The model selection is justified by the need to compare an interpretable linear baseline against a powerful non-linear ensemble method.

8.1 Model 1: Linear Regression

8.1.1 Algorithm Overview

Linear Regression models the relationship between the target variable y (log market value) and input features X as a linear combination: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_n x_n + \varepsilon$. The coefficients β are estimated by Ordinary Least Squares (OLS), which minimises the sum of squared residuals between predicted and actual values.

Configuration	Value
Algorithm	Ordinary Least Squares (OLS) Linear Regression
Library	sklearn.linear_model.LinearRegression
Regularisation	None (vanilla OLS)
Input Features	38 standardised features (marking excluded)
Training Data	4,543 samples
Random State	N/A (deterministic algorithm)
Hyperparameters	Default (fit_intercept=True, copy_X=True)

8.1.2 Strengths & Limitations

Strengths of Linear Regression in this context:

- High interpretability — coefficients directly show feature impact on log-value
- Fast training time — analytical solution, no iterative optimisation required
- No hyperparameters to tune — serves as a stable, reproducible baseline

Limitations:

- Assumes linear relationships — many skill-value interactions are non-linear
- Sensitive to multicollinearity — correlated features inflate coefficient variance
- Cannot capture interaction effects between features (e.g., age × skill level)

8.2 Model 2: Random Forest Regressor

8.2.1 Algorithm Overview

Random Forest is an ensemble learning method that builds a collection of decision trees and averages their predictions. Each tree is trained on a bootstrap sample of the data (bagging), and at each split, only a random subset of features is considered (feature randomisation). This combination reduces variance and prevents overfitting.

Configuration	Value
Algorithm	Random Forest Regressor (Bagging Ensemble)
Library	sklearn.ensemble.RandomForestRegressor
Base Estimator	CART Decision Tree (MSE criterion)
Number of Trees (default)	100 estimators
Max Features	auto (1/3 of features per split)
Bootstrapping	Enabled (with replacement)
Random State	42 (reproducibility)
Input Features	39 features (full X — unscaled, including marking indirectly)
Training Data	4,543 samples

8.2.2 Why Random Forest over Linear Regression?

Player market value is inherently non-linear. A 23-year-old with 85 ball control is valued vastly higher than expected by a simple linear extrapolation — the interaction between age, skill level, and position creates complex value surfaces. Random Forest naturally captures these interactions through tree structure, making it significantly more appropriate for this problem.

- Handles non-linearity and interaction effects without explicit feature engineering
- Robust to outliers — individual trees can specialise on different regions of the value distribution
- Built-in feature importance scores provide interpretability despite ensemble complexity
- Does not require feature scaling — distance-insensitive algorithm
- Handles the wide dynamic range of the FIFA dataset better than linear methods

8.3 Model Comparison Summary (Before Tuning)

Model	MAE	MSE	R^2	Complexity	Interpretability	Train Time
Linear Regression	0.9269	1.3321	0.7516	Low	High	Minutes
LR with PCA	0.9269	1.3321	0.7516	Low	High	Minutes
Random Forest	0.4598	0.4821	0.9101	High	Medium	~10 min

> **KEY RESULT:** Random Forest reduces MAE by 50.4% and improves R^2 from 0.7516 to 0.9101 compared to Linear Regression — a dramatic improvement that justifies the additional computational cost.

9. Hyperparameter Tuning / Model Optimization

Hyperparameter tuning is the process of systematically searching for the optimal configuration of model-specific parameters that are not learned from data. GridSearchCV was employed for exhaustive hyperparameter optimisation of the Random Forest Regressor.

9.1 GridSearchCV Methodology

GridSearchCV performs an exhaustive search over a specified parameter grid using k-fold cross-validation. For each combination of hyperparameters, it trains and evaluates the model k times on different subsets of the training data, reporting the mean cross-validation score for each combination.

GridSearchCV Setting	Value
Cross-Validation Folds	5-fold CV
Scoring Metric	R^2 (coefficient of determination)
n_estimators	[100, 200]
max_depth	[10, 20, None]
min_samples_split	[2, 5]
Total Combinations	2 × 3 × 2 = 12 configurations
Total Model Fits	12 × 5 = 60 model fits
Best n_estimators	200
Best max_depth	20
Best min_samples_split	2

9.2 Hyperparameter Explanations

Hyperparameter	Explanation & Impact
n_estimators	Number of decision trees in the forest. More trees → lower variance, better performance, but higher cost. 200 > 100 provides marginal but consistent improvement.
max_depth	Maximum depth of each tree. Controls overfitting — shallow trees underfit, unlimited depth overfits. max_depth=20 balances complexity and generalisation.
min_samples_split	Minimum samples required to split an internal node. Lower values create more complex trees. min_samples_split=2 (default) performed best, indicating the training data is dense enough to support fine-grained splits.

9.3 Grid Search Results

Configuration (Selected)	Approx. CV R^2
n_estimators=100, max_depth=10, min_samples_split=2	~0.897
n_estimators=100, max_depth=20, min_samples_split=2	~0.905
n_estimators=200, max_depth=10, min_samples_split=5	~0.894
n_estimators=200, max_depth=20, min_samples_split=2	~0.910 ✓ BEST
n_estimators=200, max_depth=None, min_samples_split=2	~0.908
n_estimators=100, max_depth=None, min_samples_split=5	~0.901

9.4 Effect of Tuning on Performance

Model	MAE	MSE	R^2 Score
Random Forest (Default)	0.4598	0.4821	0.9101
Tuned Random Forest (Best Params)	0.4584	0.4792	0.9106
Improvement	0.0014 (↓)	0.0029 (↓)	0.0005 (↑)

The improvement from tuning is modest — R^2 improves from 0.9101 to 0.9106 (a gain of 0.05%). This suggests the default Random Forest configuration is already well-suited to this dataset. The marginal gains from tuning are consistent with research showing that ensemble models are often robust to hyperparameter choices within a reasonable range.

> **INTERPRETATION:** The modest tuning improvement (0.05% R^2 gain) indicates that the default Random Forest is already near-optimal for this dataset. The primary value of the tuning exercise is methodological validation and the selection of the confirmed best configuration (n_estimators=200, max_depth=20, min_samples_split=2) for production deployment.

10. Performance Evaluation

All models were evaluated on the held-out test set (1,136 samples, 20% of the dataset) using three standard regression metrics: Mean Absolute Error (MAE), Mean Squared Error (MSE), and R^2 Score. These metrics collectively assess accuracy, error magnitude, and explanatory power.

10.1 Evaluation Metrics Explained

Metric	Interpretation	Range
MAE (Mean Absolute Error)	Average absolute difference between predicted and actual values. Robust to outliers. Lower = better.	0 to ∞
MSE (Mean Squared Error)	Average squared difference. Penalises large errors more heavily than MAE. Lower = better.	0 to ∞
R^2 Score (Coefficient of Determination)	Proportion of variance in the target explained by the model. 1.0 = perfect fit, 0 = baseline mean model, <0 = worse than baseline.	-∞ to 1.0

Note: Since the target variable is log-transformed, all error metrics are in log-scale units. An MAE of 0.46 in log space corresponds to a multiplicative factor of e^0.46 ≈ 1.58 in the original dollar scale — meaning predictions are within roughly 58% of actual values on average, which is competitive for financial/sports valuation tasks.

10.2 Comprehensive Results Table

Model	MAE	MSE	R^2 Score	Status
Linear Regression	0.9269	1.3321	0.7516	Baseline
LR with PCA (Full)	0.9269	1.3321	0.7516	Same as LR
Random Forest (Default)	0.4598	0.4821	0.9101	Strong
Tuned Random Forest	0.4584	0.4792	0.9106	Best Model ✓

10.3 Model-by-Model Analysis

10.3.1 Linear Regression Performance

Linear Regression achieved an R^2 of 0.7516, meaning it explains 75.16% of the variance in log-transformed player values. While respectable for a baseline, the MAE of 0.9269 indicates that

predictions deviate from actual log-values by nearly 1.0 unit on average. In the original price domain, this represents substantial prediction error. The model struggles particularly with outlier players whose value arises from non-linear combinations of attributes (e.g., young players with exceptional reactions).

10.3.2 Random Forest Performance

The default Random Forest (100 trees) achieves $R^2 = 0.9101$, explaining over 91% of value variance — a remarkable improvement over the linear baseline. The MAE of 0.4598 represents a 50.4% reduction in average prediction error. This model successfully captures the non-linear, interaction-driven relationships in the FIFA dataset, particularly the dominant predictive power of 'reactions' and 'ball_control'.

10.3.3 Tuned Random Forest Performance

The GridSearchCV-tuned Random Forest (200 trees, max_depth=20, min_samples_split=2) achieves $R^2 = 0.9106$ — marginally better than the default configuration. The improvement is statistically small but the tuning exercise confirms the robustness of the Random Forest approach and validates the default hyperparameters as near-optimal for this dataset.

10.4 R^2 Score Visual Comparison

Model	R^2 Score	Visual Bar
Linear Regression	0.7516	75.2%
LR with PCA	0.7516	75.2%
Random Forest	0.9101	91.0%
Tuned Random Forest	0.9106	91.1%

10.5 Generalisation Analysis

The train-test split methodology (80/20, random_state=42) ensures that performance metrics reflect true generalisation to unseen data. Several checks were performed:

- No data leakage: scaler fitted on train only, applied to test separately
- No target leakage: value column not used as a feature in any model
- Consistent random seed: all experiments used random_state=42 for reproducibility
- 5-fold CV in GridSearchCV: inner loop validation prevents overfitting to test set during tuning

The close alignment between CV scores (~0.910) and test set scores (0.9106) confirms that the model generalises well and is not overfitting to the training data.

11. Business Insights & Interpretation

The machine learning analysis yields a rich set of actionable business insights for football clubs, agents, and analytics organisations. These insights go beyond model accuracy to provide strategic implications for the football transfer market.

11.1 Feature Importance Rankings

Feature importance from the tuned Random Forest quantifies each feature's contribution to predictive accuracy. These are computed as the mean decrease in node impurity (Gini importance) across all trees:

Rank	Feature	Importance Score	% Contribution	Business Interpretation
1	reactions	0.5905	59.05%	Dominant predictor — reflects overall player quality
2	ball_control	0.1188	11.88%	Second most important technical skill
3	age	0.0498	4.98%	Prime-age players (23–29) command highest values
4	stand_tackle	0.0264	2.64%	Defensive utility adds moderate value
5	interceptions	0.0213	2.13%	Defensive awareness is market-valued
6	slide_tackle	0.0196	1.96%	Physical defensive contribution
7	dribbling	0.0187	1.87%	Attacking flair is commercially attractive
8	sprint_speed	0.0119	1.19%	Pace is a premium physical attribute
9	gk_diving	0.0110	1.10%	GK-specific — goalkeeper valuation sub-model
10	gk_reflexes	0.0110	1.10%	Confirms GK market has distinct drivers

11.2 The 'Reactions' Phenomenon

The outsized importance of 'reactions' (59.05% of predictive power) is a fascinating finding with several plausible explanations:

- In FIFA game design, 'reactions' is a meta-attribute that reflects how quickly a player processes game situations — it is a proxy for overall cognitive and physical quality.
- Top-valued players (Mbappé, Haaland, De Bruyne) all have reactions scores of 88+, making it a near-perfect proxy for elite talent tier.
- 'Reactions' correlates strongly with composite skill metrics, concentrating variance explanation in a single feature.

> **BUSINESS CAUTION:** While reactions is statistically the strongest predictor, clubs and agents should not over-rely on a single attribute. The feature's dominance in the Random Forest may reflect FIFA's game design choices rather than a pure causal relationship in the real football market. Real-world valuation should incorporate performance data, contract length, injury history, and market liquidity alongside technical attributes.

11.3 Age-Value Dynamics

Age contributes 4.98% to predictive power and reveals a nuanced non-linear relationship:

Value Quartile	Mean Age	Interpretation
Low (Bottom 25%)	24.5 years	Youngest cohort — emerging talent not yet fully valued
Mid-Low (25–50%)	26.4 years	Rising stars entering prime valuation window
Mid-High (50–75%)	26.9 years	Peak prime cohort — highest average skill ratings
High (Top 25%)	27.5 years	Established stars — experience premium + peak performance

Players in the highest value quartile average 27.5 years — slightly older than the full dataset mean of 26.3 years. This reflects the premium placed on proven, experienced talent in the prime years. The youngest cohort (bottom 25% by value, mean age 24.5) represents developing players whose potential has not yet been converted into market premium.

11.4 Goalkeeper Market Segmentation

The presence of gk_diving and gk_reflexes in the top 10 features (ranked 9th and 10th) suggests that goalkeeper valuation follows fundamentally different patterns from outfield players. The Random Forest implicitly learns these different value regimes, but future work should consider separate goalkeeper and outfield player models.

11.5 Technical vs. Physical Attributes

A clear pattern emerges from feature importance: technical and cognitive attributes (reactions, ball_control, dribbling) dominate physical attributes (strength, jumping, acceleration) in determining market value. This aligns with modern football's emphasis on technical quality and tactical intelligence over pure athleticism.

Feature	Category	Importance
Reactions	Cognitive/Composite	59.05%
Ball Control	Technical	11.88%
Age	Demographic	4.98%
Stand Tackle / Interceptions	Tactical/Defensive	4.77%
Dribbling	Technical	1.87%
Sprint Speed	Physical	1.19%
Strength / Balance	Physical	<1.0% each

12. Recommendations

Based on the modelling results, feature analysis, and business insights derived from this project, the following strategic and technical recommendations are proposed for different stakeholder groups.

12.1 Recommendations for Football Clubs

- **Prioritise players with high 'reactions' and 'ball_control' scores relative to their listed market value. These attributes have the highest predictive weight and can identify undervalued players.:** Transfer Targeting
- **Target players aged 21–24 with strong reactions and technical scores — they are at the cusp of their prime valuation window. Acquiring them early (before their value peaks at 26–28) maximises return on investment.:** Age-Value Optimisation
- **Apply separate valuation models for goalkeepers, as their attribute profiles (gk_diving, gk_reflexes) differ fundamentally from outfield players, creating different value drivers.:** Goalkeeper Segmentation
- **Use the model to identify players in the lower value quartile who have technical attribute profiles similar to mid-upper quartile players — these represent potential undervalued acquisitions.:** Moneyball Approach

12.2 Recommendations for Model Improvement

- **Re-run Linear Regression with only 10–17 PCA components (90–95% variance) to test whether dimensionality reduction improves generalisation on out-of-sample data.:** Truncated PCA
- **Replace LabelEncoder with Target Encoding for 'club' and 'country' to preserve ordinal information based on average player value per club/country, reducing false ordinality.:** Target Encoding
- **Train a dedicated model on goalkeeper records only, using GK-specific features (gk_diving, gk_kicking, gk_handling, gk_reflexes, gk_positioning) to improve GK valuation accuracy.:** Separate Goalkeeper Model
- **Evaluate XGBoost or LightGBM as alternatives to Random Forest. These gradient boosting algorithms often outperform Random Forest on tabular data and support native handling of missing values.:** Gradient Boosting
- **Implement SHAP (SHapley Additive exPlanations) for model interpretability, providing per-prediction feature attributions rather than global importance scores.:** SHAP Values

12.3 Recommendations for Data Collection

- **Supplement FIFA attributes with actual transfer fee data from Transfermarkt or similar sources to validate model predictions against real market transactions.:** Real-World Transfer Data

- **Collect multi-season data to model how player values change over time, enabling a time-series component that captures career trajectory.:** Temporal Data
- **Incorporate contract length and wage data, as players with short contracts are typically available at lower fees — a factor not captured in the current dataset.:** Contract Information
- **Add injury history as a feature, as chronic injuries significantly reduce market value regardless of attribute scores.:** Injury Records
- **Integrate real match statistics (goals, assists, pass completion, defensive actions) to complement the FIFA simulation attributes with real-world performance data.:** Performance Metrics

12.4 Deployment Recommendations

- **Deploy the tuned Random Forest as a REST API endpoint, accepting player attribute vectors and returning predicted market values with confidence intervals.:** API Service
- **Implement automated alerts when a player's predicted value significantly exceeds their current contract fee — highlighting potential transfer opportunities.:** Threshold Alerts
- **Retrain the model annually with updated FIFA data to capture market inflation, rule changes, and evolving player valuation patterns.:** Regular Retraining
- **For production deployment, average predictions from LR and RF models to reduce variance and improve stability across edge cases.:** Ensemble Averaging

13. Conclusion

This project successfully demonstrates the application of machine learning techniques to the problem of FIFA player market value prediction. Starting from raw, heterogeneous data and progressing through a rigorous pipeline of quality assessment, exploratory analysis, preprocessing, dimensionality reduction, model development, and hyperparameter tuning, the project achieves a final R^2 score of 0.9106 — meaning the model explains over 91% of the variance in player market value.

13.1 Summary of Findings

#	Finding	Detail
F1	Best Model	Tuned Random Forest (n_estimators=200, max_depth=20, min_samples_split=2) achieves R^2=0.9106, MAE=0.4584 on log-transformed values.
F2	Key Predictor	'Reactions' is the dominant feature with 59.05% importance — a composite proxy for overall player quality.
F3	Technical > Physical	Technical and cognitive attributes outweigh physical attributes in predicting market value, reflecting modern football values.
F4	Non-linearity Matters	Random Forest outperforms Linear Regression by 21.2% in R^2 (0.9101 vs 0.7516), confirming non-linear relationships in the data.
F5	PCA Finding	PCA with all components is mathematically equivalent to full-feature LR. Truncated PCA should be explored for dimensionality savings.
F6	Tuning Impact	GridSearchCV produced marginal improvement (+0.05% R^2), validating the robustness of default Random Forest configurations.
F7	Age Insight	Peak-valued players average 27.5 years; youth players (21–24) represent high-upside investment targets.

F8	Data Quality	Only 'marking' (100% null) required special handling; the dataset was otherwise clean and complete.

13.2 Limitations

- FIFA attribute ratings reflect game simulation design decisions, which may not perfectly mirror real-world scouting assessments or transfer valuations.
- LabelEncoder introduces artificial ordinality for club and country — a limitation for strict interpretability.
- The model was trained on a single season's data and may not capture year-over-year market inflation.
- No position information is included in the dataset — goalkeepers and strikers are treated uniformly, limiting position-specific accuracy.
- The 'marking' column being entirely null limits the completeness of defensive attribute representation.

13.3 Future Work

- Implement XGBoost/LightGBM for potential further accuracy gains on the same dataset.
- Explore SHAP-based model explainability to provide per-player value attribution breakdowns.
- Build separate goalkeeper and outfield player valuation sub-models.
- Integrate real-world transfer fee data from Transfermarkt to ground-truth validate predictions.
- Develop a web-based prediction interface allowing clubs to input player attributes and receive instant market value estimates with confidence intervals.

13.4 Final Statement

This project establishes a robust, reproducible, and academically rigorous machine learning framework for player valuation in professional football. The Tuned Random Forest model, achieving $R^2 = 0.9106$, represents a production-ready baseline that outperforms traditional linear methods and provides actionable feature insights. The findings validate the 'moneyball' philosophy — quantitative attribute analysis can identify market inefficiencies and inform smarter, data-driven transfer decisions in the modern football economy.

> **PROJECT OUTCOME:** The successful application of Random Forest regression to FIFA player valuation demonstrates that machine learning can meaningfully support decision-making in the multi-billion-dollar professional football transfer market. With $R^2 > 0.91$ and systematic feature importance analysis, this model is ready for pilot deployment in a real-world football analytics context.

— End of Report —